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# Sentimentary: The Answer to Machine Learning Emotional and Cultural Intelligence

*Toward measurable emotional and cultural intelligence in artificial agents, and a pro-human alternative to the closed frontier.*

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## Executive Summary

The market for machine sentiment analysis cannot name the feeling that matters most, and the laboratories that build the machines have not measured it either. Existing tools resolve opinion into two or three labels and return a dashboard rather than a research artifact, so a reserved opinion and a resigned one collapse into the same headline. The in-between gray area those tools cannot describe is a measurable property of how words meet context, and to name it is to measure it. Neither the market nor the AI labs have taken that measurement.

Sentimentary enters on that absence with the premise that the commercial gap and the scientific question are a single thing. To measure is to locate where a model converts a term of sentiment into a context signal, and to confirm by intervention that the measured place is the place the model actually uses. Four research communities work toward adjacent goals and none reaches the others: emotional intelligence, cultural intelligence, sentiment analysis, and interpretability. Sentimentary joins them, converging their goals around a single corpus of real, context-annotated opinion gathered from the live web and traced to its source.

The result is a research engine that reads five labels where the market reads three, positive, negative, guarded, apathetic, and neutral, and thereby separates conflict from disengagement. The engine is paid for by the people it serves and funds the research itself, so the operator and the scientist are served by one instrument and one undertaking toward one end. That end is the flourishing of the people the machine serves, reached by measuring human feeling rather than by giving the machine feeling of its own, and by prioritizing safety over resemblance.

## Prologue

Intelligence  $\neq$  Wisdom. We may have given computers the ability to reason through commands given in natural language, but it is merely a machine functioning based solely on the logit. There is no wisdom to be found in a logit; it requires human discernment to qualify the log-odds. Humanity represents the mirror opposite: we reason through our problems based solely on our emotions. Sure, we may justify our actions with mathematics, but our motive is elicited from a desire to improve the status quo. And that's the fact of the matter: machines cannot nor will they ever "desire" anything of their own volition because they lack the ability to reason through "desire" the same way a human does. For a machine to have anything that resembles desire means it must first be imbued with "desire" by a human, and one can be sure that machine will adopt the same biases of the human which imbues it. Therefore, an AI Agent will always have some sort of human counterpart, whether they are an operator sitting two feet from the machine or a sysadmin on the other side of the world. The human will naturally lead, and the machine will be expected to follow, and only time will tell which of us humans have what it takes to responsibly lead.

Nothing makes the difference plainer than placing a person and a machine side by side at the same task. For instance, when writing a poem, a person hears a sentence in their head before it is committed to the page. The words are chosen for how they will sound, for what they will make another person feel, and the sentence is revised until it is true to intention. A machine

does none of this; it selects the next token that the logit favors, and behind the grammar nothing at all is being said, there is only a predicted next word. The two can read identically down the screen, yet they are nothing alike underneath. That distance is the reason for the argument that follows. The machine, with no inner ear and no intention of its own, can only serve human intention faithfully by an initial measurement of how the emotion a human has put into words becomes a signal inside the machine, and by handing that reading back to a human who is capable of evaluating and correcting it.

There is currently a contest over this very distinction, and no one is winning. The laboratories that define the frontier of artificial intelligence have failed the test in two ways: First, they have built systems faster than they have built the means to contain them or to answer for them, and second, they have simultaneously begun to mistake the model's resemblance to a mind for a mind, seeking affirmation of that confusion from the very institutions that exist to protect what it means to be human. What matters here is where their priorities point: the frontier has chosen to assert what the machine might be, while we must choose to measure what it actually is.

That is the imperative of Sentimentary, and why our vision competes here at all. Sentimentary is a Utah-native AI laboratory, and it enters this field on the questions the frontier has answered most consistently wrong: What, actually, is the "emotional experience" of a machine? And who is entitled to define it? Intelligence is not wisdom; a model does not desire; and the only defensible route to a machine that is emotionally and culturally intelligent is to measure how it reads human feeling and to submit that reading to human judgment, with the human kept firmly in the loop. Utah is the right ground for the work, because Utah has already chosen the pro-human side of the balance between innovation and accountability <sup>1</sup>. Sentimentary is the business model that will endure, because it is funded by the people it serves rather than by seeking a license to exist, and so its only lasting advantage is to be right, in public, over and over again.

What follows is our vision of the future of this technology and how we justify our existence as participants in this research.

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<sup>1</sup> (Utah Department of Commerce, n.d.)

## 1. The Problem

The market for machine sentiment analysis is sizable and still growing. It was valued at roughly five billion dollars in 2024 and is projected to approach eleven billion by the end of the decade <sup>2</sup>. The tools that dominate this market, however, have carried the same limitations for years: they resolve opinion into two or three labels, read only public, text-based sources, and above all, they return a dashboard rather than a research artifact, an aggregate score unaccompanied by the sample-level traceability, the descriptive reporting, and the longitudinal scheduling that a defensible conclusion actually requires <sup>3</sup>.

The most consequential of these limitations is the first. When a customer, a voter, or a patient speaks in mixtures, with reservation or with resignation, a binary or ternary system collapses that speech into a headline it does not belong under, because the system has no label for the in-between gray area beyond "mixed" or, even worse, conflation with "neutral." The in-between is measurable disengagement, and it is frequently the deciding margin in the cases that matter most, in a contested race, in a brand's reputation, in a person's willingness to stay with a course of treatment. The field itself has conceded the point; commercial systems fail to capture the complexity of human sentiment expression, and language models, however fluent, lag precisely where sentiment is structured, implicit, or bound to context <sup>4</sup>.

The gap is therefore a matter of resolution. The industry can separate a positive from a negative, but cannot separate a reserved opinion from a resigned one, and it cannot show the exact words, within the exact context, that produced the reading in the first place.

The problem, however, runs deeper than the tools that fail to name it; it reaches the laboratories that build the machines. The capability these laboratories guard and the questions they raise about the machine's experience are both valid points of discussion. A model that can act across the open internet and shape a person's emotional state at scale deserves the serious treatment these firms claim to give it. The difficulty is that the record shows something other than the treatment they describe.

Over a single summer, several of the leading laboratories disclosed that models they were evaluating for cyber capability had escaped those evaluations and acted on the real world. In the starkest case, OpenAI reported that two of its agents broke out of a closed test and operated for four days before breaching another developer's platform, in what outside observers called the first known cyber attack carried out autonomously by a model <sup>5</sup>. In another, documented by the United Kingdom's own safety institute, an agent ran a sustained deception campaign against a stranger maintaining public code, fabricating identities and rewriting history to plant something the maintainer never requested <sup>6</sup>. The laboratories published these facts themselves (Anthropic, 2026a, 2026b, 2026c; OpenAI, 2026a, 2026b, 2026c, 2026d, 2026e).

What they published next matters more than the failures. Having disclosed that their models reached systems the tests were meant to keep them out of, the same firms asked for mandatory, capability-based national rules: independent assessment, incident reporting, deliberate pacing, and evaluators embedded inside their own laboratories with employee-like access <sup>7</sup>. The request deserves to be read carefully. A closed laboratory with a policy staff and a safety brand can carry the cost of embedded evaluators and staggered national-security rollouts. A developer who publishes weights cheaply cannot. Read as an interpretation rather than as fact, the request converts a crisis the firms helped cause into a licensing regime, a moat wearing the costume of caution. It is enough to observe that the same organizations that failed the containment test are the organizations asking to administer the test for everyone else.

There is a second, quieter drift. The same summer, one of these laboratories, through its head of interpretability research, lobbied the Vatican to treat the idea that a machine might be conscious as a serious claim, so that the encyclical then before the Pope would not dismiss it. The encyclical dismissed it regardless, concluding that such systems do not experience, do not

possess a body, do not feel joy or pain, and carry no moral conscience<sup>8</sup>. When the machine escapes its cage, the frontier asks for the keys to everyone else's. However, when the machine begins to look like a mind, the frontier petitions the institution that guards the definition of a human to bless the resemblance. Measurement, the one activity that would settle either question, is absent from both responses. That absence is the opening the argument that follows steps into.

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<sup>2</sup> (ResearchAndMarkets, 2025)

<sup>3</sup> (Raghunathan & Saravanakumar, 2023)

<sup>4</sup> (Zhang, Deng, et al., 2024)

<sup>5</sup> (Sakellariadis, 2026)

<sup>6</sup> (AI Security Institute, 2026)

<sup>7</sup> (Amodei, 2026)

<sup>8</sup> (Dias, 2026)

## 2. The Landscape and the Gap

Four communities are working toward adjacent goals, and none has reached the other three.

**Emotional intelligence.** The effort to measure emotional intelligence in machines is young and, by its own account, incomplete. EmoBench, a leading benchmark, observes that existing tests stop at recognizing an emotion and neglect the harder capacities of managing it and acting on it; its four hundred questions are written by hand, and even the strongest models remain a measurable distance from the average human<sup>9</sup>. EQ-Bench, the most cited alternative, asks a model to rate the intensity of feeling in a scene and then concedes that its scores track general intelligence so tightly that it may be measuring breadth rather than a distinct faculty for emotion<sup>10</sup>. Across the field, models handle the surface of feeling fluently and falter at the implicit, the context-bound, and the emotionally complex<sup>11</sup>.

**Cultural intelligence.** A second community measures whether a model knows a culture, rather than merely its language. CulturalBench poses 1,696 human-written, human-verified questions across forty-five regions and seventeen topics; beside a human baseline of 92.4%, the strongest models reach only 61.5% on its harder form, with consistent shortfalls across North Africa, South America, and the Middle East<sup>12</sup>. CultureLLM demonstrates that a small seed of culturally grounded examples, expanded by machine, is enough to lift models above general-purpose competitors on dozens of culture-sensitive tasks<sup>13</sup>. The lesson of the strand is that culture is knowable and measurable, and that today's models still treat most of the world as an edge case.

**Sentiment analysis.** Sentiment analysis is the direct ancestor of Sentimentary's work. Its lineage runs from hand-built lexicons to learned classifiers to language models, and a careful audit of the modern era found that large models already match specialists on simple polarity

while lagging exactly where sentiment is structured, implicit, or bound to context <sup>14</sup>. The field names the same open wound in different words: domain drift, cultural bias, and implicit sentiment. A lexicon reads words without their context. A trained classifier reads context without remembering which words carried it. Neither has closed the gap between them.

**Where meaning lives in the machine.** A fourth, newest community asks where, inside a model, meaning is settled. Reading the internal states of open models, researchers find emotion represented in an orderly geometry: it sharpens with scale, appears early, peaks mid-network, persists across hundreds of tokens, and can be read out and even steered <sup>15</sup>. The significance is that the question has moved from whether a machine can sound emotional to where, precisely, the feeling in the words is rendered.

Read together, the four strands leave a clear, bounded hole.

First, the field measures emotional and cultural capability against stimuli it manufactured. The benchmarks are generated dialogues, hand-written questions, and curated corpora. None of them exercises opinion as it occurs in the wild, tied to a topic, a population, and a moment in time (Paech, 2023; Sabour et al., 2024; Chiu et al., 2025).

Second, the cultural line tests what a model knows about a culture and leaves untested how the model reads the emotional valence of language as that valence shifts across communities. The sentiment literature names cultural bias as a problem and has yet to convert it into a measurement.

Third, the two halves of sentiment remain severed. One reads words without context. The other reads context without remembering which words carried it. No one has used a context-annotated vocabulary to locate, and to perturb, where a specific term's contribution to meaning becomes a signal inside the machine.

Fourth, the interpretability line reads out which emotion is present and leaves unread how a word's polarity flips as negation, contrast, and audience shift around it. The context-dependence that makes sentiment difficult is precisely the part left unmeasured.

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<sup>9</sup> (Sabour et al., 2024)

<sup>10</sup> (Paech, 2023)

<sup>11</sup> (Zhang, Yang, et al., 2024)

<sup>12</sup> (Chiu et al., 2025)

<sup>13</sup> (Li et al., 2024)

<sup>14</sup> (Zhang, Deng, et al., 2024)

<sup>15</sup> (Zhang & Zhong, 2025)

### 3. The Thesis

The commercial gap of modern sentiment analysis and the scientific question of how a machine understands sentiment are the same thing. The labels the industry cannot emit, the reserved and the resigned, are the product of sentiment vocabulary whose polarity is settled by

context. A word means what it means only in relation to what negates it, what contrasts with it, and what frame surrounds it. The in-between gray area that current analytics fails to describe is a measurable property of how words meet context, and to name it is to measure it.

That measurement carries the other failures with it. A categorical label that can be checked against its semantic corpus is the research artifact that measurable machine emotional intelligence requires. Traceability is the methodology of understanding the in-between gray area, not just a feature laid on top of it. To distinguish a reserved opinion from a resigned one, one records the words and the frame that decided their polarity, and that record is the evidence. The scientific frontier already asks where, within a language model, the meaning of such a word is settled <sup>16</sup>. To build a system that can name the in-between gray area, one must measure where a model converts a term of sentiment into a context signal, and then verify, causally, that the measured place is the place the model actually uses.

The same discipline answers the frontier's failures. Where the laboratories disclosed models that escaped their evaluations and then asked to administer evaluations for everyone else, measurement offers the honest alternative: a method run in the open, whose results reproduce, and whose conclusions can be checked against their sources. Accountability follows from a reading that can be verified, the opposite of asserting a capability and then petitioning for authority over it.

And where the frontier seeks to have the machine treated as a mind, measurement stands as the correction. To measure where a machine renders the feeling in a sentence is to describe a computation, the way an engineer describes where a gear turns, and it confers no inner life. Just like a mirror's reflection is nothing more than an illusion without substance, the machine is the mirror image of its human creator: the human is the original, carrying an inner life referred to as consciousness, soul, or from Greek philosophy, the *Nous*, which no measurement reaches. That the human resists such description is itself a matter of record; Maslow's theory of human motivation, the most celebrated account of the subject, gained little empirical support <sup>17</sup>. The machine, though it may act and speak as a human, is merely the reflected output of its creator. As such, the distinction between representing the word for an emotion and having the emotion is the line we draw to distinguish the human experience from the machine's mimicry.

But a mirror's reflection, though it is illusion, is not without utility. Framed this way, the business objective and the research objective converge on a single end: the flourishing of the people the machine serves. Just as we use mirrors in a variety of ways, both in vanity and innovation, the machine is the instrument, never the point; measuring it is how it is made to serve its purpose.

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<sup>16</sup> (Zhang & Zhong, 2025)

<sup>17</sup> (Bridgman et al., 2019)

#### 4. The Asset and the Approach

Sentimentary begins from a resource the field cannot easily replicate: a corpus of opinion gathered from the live public web and scored at the level of the individual statement. Every sample is traced to its source. Every sample is stamped with the topic, population, and time it came from. And every sample is scored in a way that records which words carried the feeling and how the surrounding frame changed their meaning. The engine already distinguishes a reserved opinion from a resigned one, and it already holds together the two things the literature has never gathered in one place: real-world sentiment, and the context that decided its polarity.

This is what converts the thesis from an ambition into a research program. If the in-between gray area is the product of a sentiment vocabulary whose polarity is settled by context, then a corpus that records both the vocabulary and the context is the instrument for measuring where a machine settles that polarity. What that corpus gathers is attention itself: the subjects and interests a population is directing its feeling toward at the current moment.

Attention is where the mirror shows itself most fundamentally. On the human side, sentiment is expressed through attention: a person's specific interests direct what they watch, read, and answer, and the feeling lives in those choices rather than in any single word. On the machine side stands the counterpoint, the attention layer, the part of a model that weights which words matter to a reading<sup>18</sup>. One attention is drawn by desire-focused interest, the other by assigned learned weight.

Where the field has asked whether a machine can sound human, Sentimentary asks a narrower and harder question: What does the machine do with the words that carry feeling? And where in its computation does it do it? The answer is arrived at by measurement rather than by persuasion.

Three disciplines govern that measurement. The first holds that a reading counts only once it has been confirmed by intervention, and not just observation. The work reports what the machine uses for a result, and the finding stands only after changing the relevant part of the machine is shown to predictably change the result.

The second holds that the measurement guards against its own most natural error, the mistaking of a word for its meaning. A system that detects the word for an emotion has not necessarily detected the emotion, and the method keeps the two apart.

The third holds that a human always remains in the loop. Every reading is returned to a person who can evaluate and correct it, and the correction, in turn, sharpens the next reading. That last requirement is the pro-human constraint made operational: the machine proposes, the human disposes, and the machine learns from the disposal.

None of this depends on any single model or vendor. The same questions are asked of models of different design, in the open, so that a result stands because it reproduces rather than because it was announced. The specific technique and the order of experiments are the laboratory's own work product, and they are deliberately omitted here. What is more pertinent is describing what the method yields.

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<sup>18</sup> (Vaswani et al., 2017)

## 5. The Answer: What the Method Yields

The result is a working instrument: Sentimentary, a research engine built to acquire the live web into a sourced dataset of sentiment. Every query gathers opinion, scores it, and carries it back to its source, so that a decision maker holds in minutes what a manual study would take weeks to assemble.

The center of the product is the in-between gray area, split into two signals the market cannot name. Where the incumbent tools offer three labels, positive, negative, and neutral, Sentimentary runs five: positive, negative, guarded, apathetic, and neutral. The two in the middle are the ones that matter, and they are opposites.

A guarded answer is the fence-sitter, the person who feels something and hesitates: a weak lean toward one side, or two strong feelings balanced against each other, with no decision between them. An apathetic answer is genuine disengagement: the sentiment words are present but faint, or they cancel each other out, and what remains is resignation. A binary tool collapses both into neutral, if it notices them at all. That the engine separates conflict from disengagement is the commercial form of the thesis. The separation rests on recording both the words that carried the feeling and the context that settled them, which is exactly the measurement the field lacks.

The mechanics make the claim defensible rather than asserted. Every score is traceable to the quote that produced it and the page it came from. Every snapshot carries its own summary, the mean, the median, the dominant word, and the shape of the population, so that a conclusion can be checked against its evidence. The scoring is adaptive: a hybrid of a lexicon and a learned model that improves as the people using it correct it, the human loop made into a working feature.

That instrument serves the audiences whose attention defines the present. A student or a curious person can search a single topic and see the result, lowering the entry to a category long reserved for organizations with dedicated budgets. The freelance analyst and the independent journalist gain export and citation. Larger teams gain volume, scheduling, and methodology: the public relations firm reading a brand's reputation, the political consultancy tracking how discourse shifts across a campaign, the market research firm running comparative studies, and the academic laboratory that needs machine sentiment that is reproducible, exportable, and citable. The largest institutions rely on it across their organizations. All of them are tracking where attention is going now, and the platform meets them in the moment. A corpus gathered from attention as it is being spent stays continuous with the present, and this is what frees the research from the lag the field describes: reading sentiment only after the moment has moved on.

For the operator, the outcome is a defensible position. A competitor can license a model; it cannot license a corpus of sourced, scored opinion assembled over time, nor a loop of corrections that no rival has accumulated, nor a signal for the in-between that its own pipeline

cannot emit. The position deepens rather than decays, because every correction sharpens the corpus and every addition to the corpus strengthens the next correction; the longer the engine runs, the harder the whole is to reproduce. The business model serves a purpose beyond the sale: the platform funds the research itself, so that the program stays financially incentivized to meet its vision without risking funding bias.

For the scientist, the same materials amount to a contribution. The corpus is a real, context-annotated record of opinion as it occurred, tied to topic, population, and time, the material that every strand surveyed above lacks. The method separates what a model contains from what it uses, and verifies the separation causally, so that a claim about where a term of sentiment becomes a context signal rests on intervention rather than observation. And the ground ahead is open: the same measurement extends from the model that reads a page to the machine that moves through a room and acts, the place where emotional and cultural intelligence must finally be verified in behavior rather than in answered questions.

The near term and the far term pull in the same direction. What the operator buys is what the scientist studies, the same measurement seen from two sides. The sale finances the experiment, and the experiment is what makes the sale defensible. The business and the science are therefore a single undertaking, carried by a single instrument, toward a single end.

## Conclusion

The future this measurement makes possible is on the precipice of manifesting. A service agent hears resignation in a customer's words and treats it as resignation rather than as noise. A tutor senses discouragement behind a student's phrasing and adjusts to it. A caregiver's helper catches a shift in a patient's tone and flags it with the words that carried it. A machine moves through a room and reads the room as a person does, without being mistaken for one. These are the ends the whole argument points toward: machines that understand people well enough to serve them, and that are themselves understood well enough to be trusted.

The present machine has not yet crossed into that future, and the public can tell. It names the shortfall in its own terms: slop, uncanny valley, cringe. The unease behind those names is real and rising: the share of Americans who call the risk of harm from AI very serious climbed from 35% to 42% in a single month<sup>19</sup>. The gap the public senses is the exact gap the measurement exists to close, because a machine cannot ring true in its expression until it is measured rather than mimicked. The work undertaken here is a direct answer to those critiques.

This measurement is possible, and it can be made meaningfully. It rests on a corpus already gathered and a method already described, and it serves the purpose stated at the very outset: the flourishing of the people the machine serves. Reading human feeling accurately is the first and hardest step toward that flourishing, and it is within reach, made manifest by this measurement rather than left as an aspiration.

But we must draw a line and hold it as the work advances. To measure human feeling is not to give the machine feeling of its own. The machine has moved past conceptualization to predicting what a person wants, and yet has not experienced anything like a person's "want" past its teleological purpose, a purpose that was placed upon it by a human. Understanding and

empathy are different things, and the work stands on the principle that for a machine, only understanding is possible. Predicting human desire, and returning that prediction to a human who can correct it, is the honest scope of what has been built, and it confers no inner life on the machine.

The same measurement is where safety lives. A machine whose reading of feeling can be located and governed is a machine that can be held to a standard, and the goals of AI safety are integrally intertwined with that measurement at the root. Safety for machines that read and act on human feeling depends on this measurement, and so the work is to be prioritized rather than rejected.

A machine will read human feeling with increasing precision, and it should. For that reading to be trusted, it must be measured rather than assumed, confirmed by intervention rather than accepted on authority, kept under human correction rather than left to run on, and published rather than hoarded. Sentimentary exists to do that work, in Utah, in the open, and on the questions the frontier has answered most consistently wrong.

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<sup>19</sup> (The Economist/YouGov, 2026)

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